

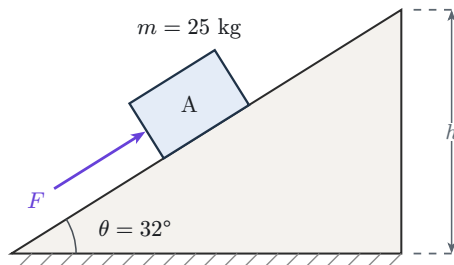
## PROBLEM 1

10 marks

**Crate on a rough incline**

A crate of mass  $m = 25 \text{ kg}$  rests on a ramp inclined at  $\theta = 32^\circ$  to the horizontal. The coefficients of friction between the crate and the ramp are  $\mu_s = 0.45$  and  $\mu_k = 0.35$ . A force  $F = 250 \text{ N}$  is applied to the crate parallel to the ramp, directed up the slope.

- Draw the free-body diagram of the crate and resolve its weight into components along and normal to the ramp.
- Show that the crate cannot remain at rest, and state the direction in which it moves.
- Find the normal force and the acceleration of the crate.
- The crate starts from rest at the foot of the ramp. Find its speed on reaching the top, a height  $h = 3.4 \text{ m}$  above the ground.



Take  $g = 9.81 \text{ m/s}^2$ .

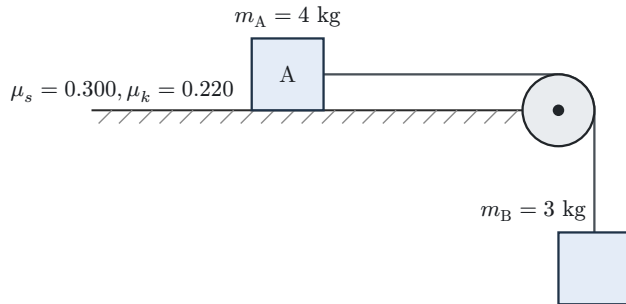
## PROBLEM 2

12 marks

**Block, cord, and hanging mass**

Block  $A$  lies on a horizontal table. A light inextensible cord runs from  $A$ , passes over a frictionless pulley at the edge of the table, and carries block  $B$ . The coefficients of friction between  $A$  and the table are given in the figure; the pulley is light and turns freely.

- Draw a separate free-body diagram for each block, and state the relation between their accelerations.
- Determine whether the blocks move when released from rest.
- Find the acceleration of the blocks and the tension in the cord.
- Block  $B$  falls a distance  $d = 1.2\text{ m}$  to the floor. Find the speed of  $A$  at that instant, and the further distance  $A$  slides before stopping.



Take  $g = 9.81\text{ m/s}^2$ .

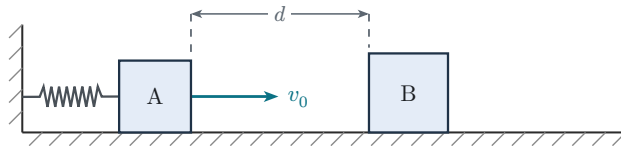
## PROBLEM 3

12 marks

**Spring launch and collision**

Block  $A$ , of mass  $2.0\text{ kg}$ , is held against a spring of stiffness  $k = 800\text{ N/m}$  compressed  $x = 0.15\text{ m}$  from its natural length, and is then released. The spring is light and the track is smooth. Block  $B$ , of mass  $3.0\text{ kg}$ , stands at rest a distance  $d = 2.5\text{ m}$  ahead of  $A$ .

- Find the speed  $v_0$  with which  $A$  leaves the spring.
- The blocks stick together on impact. Find their common velocity and the kinetic energy lost in the collision.
- Find the velocity of each block instead if the collision is perfectly elastic.
- Beyond  $B$  the track is rough, with  $\mu_k = 0.20$ . Taking the collision of part (b), find how far the blocks travel before coming to rest.



Take  $g = 9.81\text{ m/s}^2$ .

## PROBLEM 4

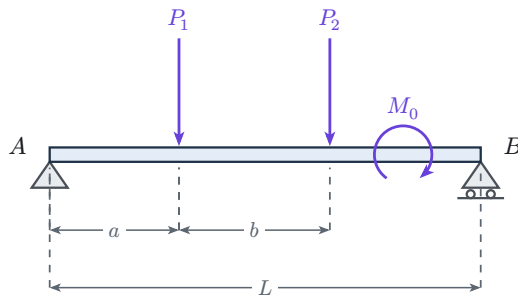
10 marks

**Simply supported beam**

A light beam  $AB$  of length  $L = 6.0\text{ m}$  rests on a pin support at  $A$  and a roller at  $B$ . It carries vertical loads  $P_1 = 4.0\text{ kN}$  and  $P_2 = 6.0\text{ kN}$  at the distances shown, with  $a = 1.8\text{ m}$  and  $b = 2.1\text{ m}$ , together with a couple  $M_0 = 5.0\text{ kN} \cdot \text{m}$ .

- Draw the free-body diagram of the beam, showing the reaction components at each support.
- Find the reactions at  $A$  and at  $B$ .
- Explain why the point of application of  $M_0$  along the beam does not affect either reaction.
- Find the distance from  $A$  at which  $P_2$  would have to act for the two reactions to be equal.

Take  $g = 9.81\text{ m/s}^2$ .



## PROBLEM 5

10 marks

**Vertical circular loop**

A small body of mass  $m = 0.40$  kg slides along a smooth horizontal track and runs onto the inside of a vertical circular loop of radius  $R = 2.0$  m. Friction is negligible throughout, and the body may be treated as a particle.

- (a) Draw the free-body diagram of the body at the highest point of the loop, and write Newton's second law along the radial direction there.
- (b) Find the least speed  $v$  at the highest point for which the body stays in contact with the track.
- (c) Find the speed the body must have on entering the loop at the bottom in order to pass the highest point at that least speed.
- (d) Find the speed at the highest point for which the track pushes on the body with a force equal to the body's weight.

Take  $g = 9.81$  m/s<sup>2</sup>.

